Harnessing the Power of Curated Metadata: Driving Scientific Discoveries Using Genomes Online Database (GOLD)

Van Handke, Anuha Reddy, Supratim Mukherjee, Galina Ovchinnikova, Dimitrios Stamatis, Cindy Li & T.B.K. Reddy

DOE Joint Genome Institute, Lawrence Berkeley National Laboratory, Berkeley, CA 94720

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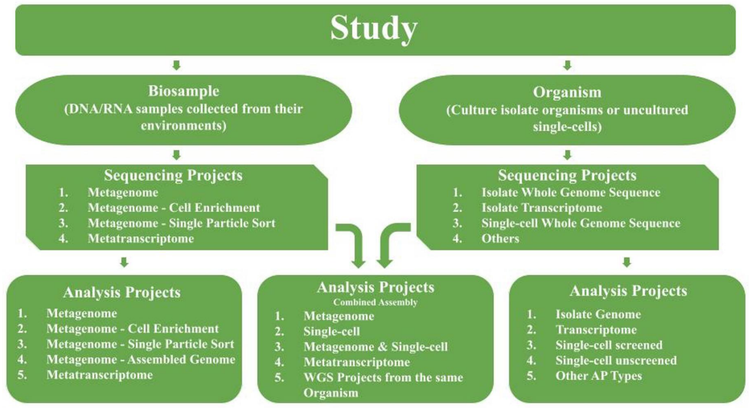
Introduction

The Genomes OnLine Database (GOLD) is a critical resource for researchers aiming to harness the power of curated metadata to drive scientific discoveries. GOLD offers an extensive repository of metadata for genome and metagenome sequencing projects, encompassing a broad range of data, including organismal taxonomy, sequencing technologies, project goals, and environmental contexts. This rich metadata enables researchers to conduct targeted queries, facilitating the identification of patterns and relationships across diverse datasets.

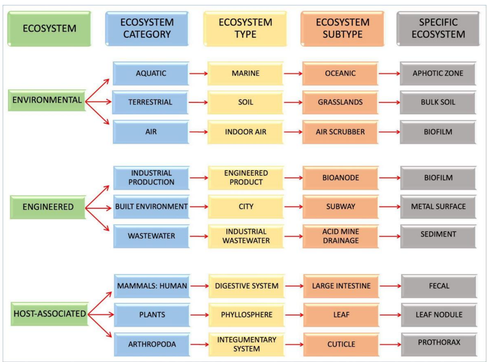
GOLD Overview

A wide range of projects along with their metadata make their way into GOLD from multiple sources. These encompass internal projects sequenced at Joint Genome Institute (JGI), user-submitted external projects, and projects acquired from public databases such as NCBI. The semi-automated curation process in GOLD involves hands-on supervision, ensuring the data's precision and dependability. The database arranges information systematically into a well-structured four-level system, making it effortless for users to navigate and explore the metadata.



Ecosystem Classification

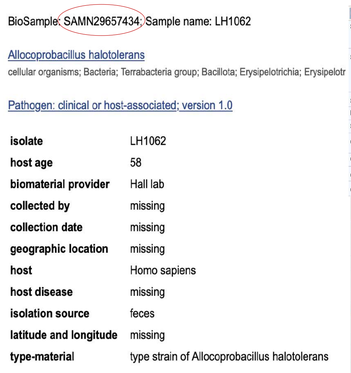
GOLD's 5-level Ecosystem classification system consists of controlled vocabulary terms spread across ecosystem, ecosystem category, ecosystem type, ecosystem subtype and specific ecosystem categories.



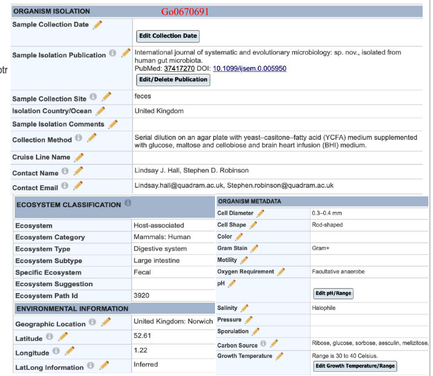
Why is Metadata Important?

For comparative analysis and hypothesis testing, it is extremely important to assure the quality of metadata for genome and metagenome projects.

For example, compare the two pages from NCBI and GOLD.



NCBI



GOLD

The NCBI page on the left is relatively empty, with many "missing" values

However, the GOLD page on the right, has a large list of manually curated metadata taken directly from a combination of NCBI BioSample and the original isolation/genome publication of the organism.

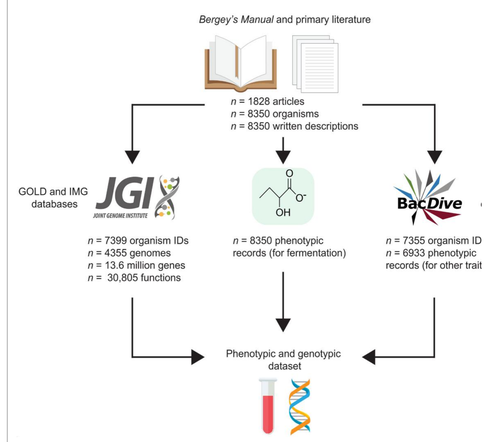
The capacity to do comparative analyses is hampered by missing or misleading metadata, which has a negative impact on the interpretation of results. Through using GOLD's comprehensive set of metadata, analyses are much more effective due to its more concise and descriptive information to choose from.

How GOLD Can be Useful to You

As one of the "core bioinformatics pipelines/web applications" (*Schriml et. al, Nature Scientific Data 2020*), GOLD's key strengths lies in its ability to support metadata-driven discoveries. This is through GOLD's hosting of hundreds of thousands of projects and associated metadata, including over 516,000 Organisms and 209, 000 Biosamples. Additionally, GOLD's integration with other bioinformatics tools and resources further enhances its utility, allowing researchers to cross-reference data and generate new hypotheses.

This access to curated metadata allows for the facilitation and exploration of genetic diversity, understanding of evolutionary processes, and the development of new biotechnological applications, demonstrating the indispensable role of GOLD in modern genomic research.

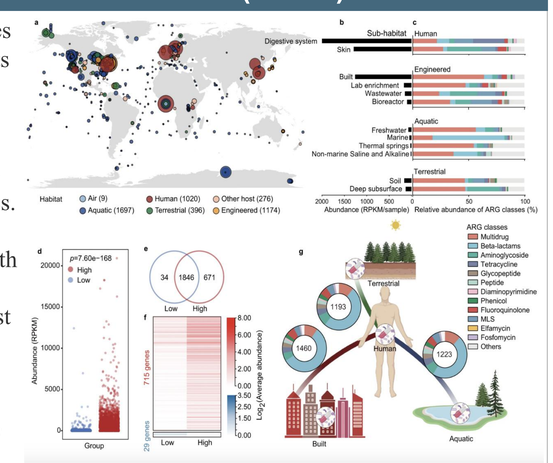
Demonstrate Complexity of Fermentation in Prokaryotes



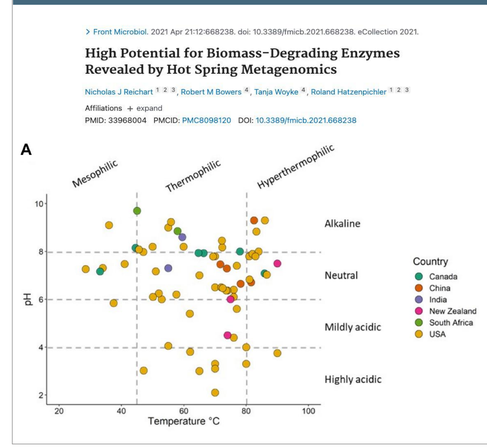
Authors used 8350 organisms (type strains) and 29 metadata categories to create a phenotypic and genotypic dataset. This led to reveal that "fermentation is both widespread (in ~30% of prokaryotes) and complex showing gaps in our knowledge" in contrast to earlier studies showing that fermentation is well understood (*Hackmann and Zhang, Science Advances 2023*)

Assess the Global Health Risk of Antibiotic Resistance Genes (ARGs)

Analyzed metagenomes from 6 types of habitats (4572 samples) that detected 2561 ARGs that collectively conferred resistance to 24 classes of antibiotics. Created a new understanding the health risks that "...will manage one of the most important threats to human and animal health" (*Zhang, et al., Nature Communications 2022*)



Discover Biomass Degrading Microbes from Hot Springs



Authors analyzed 71 metagenomic datasets from globally distributed terrestrial geothermal features to detect a wide diversity of microbes unique to varying temperature and pH ranges. Thermal sites across a wide spectrum of temperature and pH provided a unique opportunity to expand the repertoire of known extremozymes with lignocellulolytic functions.

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